

SCHOOL OF BUSINESS, ECONOMICS AND LAW

BACHELOR IN BUSINESS ADMINISTRATION (ENGLISH)

COURSE: Search, Processing and
Use of Information

YEAR: 1

TYPE: Compulsory

DEPARTMENT: Quantitative Methods

CODE: G1

SEMESTER: 1

ECTS CREDITS: 6

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1 COURSE DESCRIPTION AND MAIN OBJECTIVES

This course focuses on identifying the general and specific information sources that are most relevant to the degree and selecting the most appropriate for each task. It analyses the reliability and quality of the information and its sources, and participants learn to create and manage citations and bibliographic references, and use spreadsheets, database tables, and graphing and visualisation sheets for quantitative database management.

2 LEARNING OUTCOMES

2.1 KNOWLEDGE OR CONTENTS

C15 - Be familiar with the most relevant digital transformation tools.

2.2 SKILLS

H7 - Identify, use and analyze relevant economic information sources.

2.3 COMPETENCIES

CO1 - The ability to organize and plan their work, with self-discipline, self-demand, and rigor in their academic work related to the field of management.

CO5 - Ability to gather and interpret updated and relevant data from an economic-business perspective for making informed decisions, adding value to the management of an organization, and including a critical reflection on business ethics and their implications for society.

CO6 - The ability to work in teams, actively participating in projects and tasks in the field of business administration and management.

3 COURSE DETAILS

3.1 PROGRAM

1. From Data to Information to Intelligence: The journey from raw data to actionable intelligence together with a high-level view of the data analysis process.

2. Building Data Teams: Agile methodology and how it can help delivering valuable business insights, with particular focus on Kanban.

3. Data Collection: Informational strategic needs shape data analysis. What we are gathering data for will influence both the type of data to be collected and the methods we use to collect it. We examine the different types of data we can collect, and when to use one over the others.

4. Ethical uses of Data: The potential problems and pitfalls of using data: from cognitive biases to data biases to privacy issues.

5. Types of Data Analytics: The 4 types of Data Analytics: Descriptive Analytics, Diagnostic Analytics, Predictive Analytics and Prescriptive Analytics.

6. Exploratory Data Analysis and Descriptive Analytics: Using Descriptive Analytics to explore and describe data, including descriptive statistics and graphs.

7. Excel: Examine spreadsheet concepts and explore the Microsoft Office Excel environment. Create, open, and view a workbook. Modify a worksheet and workbook. Work with cell references. Learn to use functions and formulas. Create and edit charts and graphics. Filter and sort table data. Work with pivot tables and charts. Finding and Referencing Data inside Excel. Import and export data. Data Validation and Deployment. Use of Add-ins for advanced Data Analysis.

3.2 TRAINING ACTIVITIES

AF1 - Lectures. Professors present the theoretical contents of the course with or without student participation. 100% face-to-face.

AF2 - Practice sessions. Under the instructor's supervision, students apply the knowledge acquired in class. These sessions include workshops, case studies, problem-solving activities, hands-on labs and simulations, use of specific software, etc.). 100% face-to-face.

AF3 - Projects (individual or group). Preparation of individual or group projects, including papers, oral presentations, reports, etc. Face-to-face attendance can be 0% if the professor offers guidance during the tutoring sessions.

AF4 - Tutoring sessions (individual or group). Meetings between the professor and students, individually or in small groups, to offer personalised academic guidance. This includes solving doubts, supervising projects, preparing presentations, etc. Face-to-face attendance does not have to be 100% if there is, for example, tutoring by email.

AF5 - Individual study and work. Time spent by students working independently on each course, completing coursework, studying, searching for information, analysing data, preparing documents, etc. 0% face-to-face attendance.

AF6 - Assessment. Written, oral and practice tests, coursework, assignments, etc. that enable professors to assess whether students have achieved the desired learning outcomes. 100% face-to-face.

3.3 LEARNING METHODOLOGIES

M1 - Traditional methodology (Lectures). The instructor plays a central role in the learning process, relaying contents and mediating between students and the contents they must master. During lectures, professors can use different support technologies and carry out learning activities to encourage analysis, reflection and discussion.

M2 - Problem-based learning. Instructors pose a problem (or set of problems) that students must solve, applying solutions based on the knowledge and skills acquired in class.

M3 - Challenge-based learning. Instructors present a problematic situation that connects with real-life professional scenarios and represents a challenge. Students have to look for a solution, proposing alternatives. This methodology allows them to deepen their insights and apply them to the business world, beyond the context of the university.

M4 - Cooperative learning. Students work in small teams to carry out different tasks.

3.4 ASSESSMENT CRITERIA

3.4.1 ASSESSMENT GUIDELINES

Attendance is mandatory during the first enrollment, except for final-year courses of the degree program. Attendance below 80% results in the loss of the right to take the ordinary exam call.

Course assessment consists of continuous assessment during the class period (mandatory for all students, including retakers and exchange students, and in all academic years), and a final exam. Assignments or midterm exams are not exempting. The final exam is common to all students and set by the course faculty, covers all the contents of the syllabus, and requires a minimum grade of 5 out of 10 to pass. The official grade is recorded on a 10-point scale.

If the student fails the final exam, the official grade will be the exam grade.

If the student scores 5 or higher on the final exam, the official grade will be a weighted average of the continuous assessment (40%) and the final exam (60%), which may still result in a failing grade.

If the student does not take the final exam for any reason, the official grade will be the result of applying the 40% weighting to the continuous assessment.

If the student has not completed any assessments (neither continuous assessment nor the final exam), the official grade will be "Not Attended" (No Presentado).

In the extraordinary exam call, the grade corresponding to continuous assessment may be replaced -when it benefits the student- by the result of the weighted average between the continuous assessment grade (40%) and the grade of the final exam in the ordinary exam call (60%).

ASSESSMENT ACTIVITIES 1	
Weight:	5%
Description:	Weekly Quizzes
ASSESSMENT ACTIVITIES 3	
Weight:	5%
Description:	Theory Midterm 2
ASSESSMENT ACTIVITIES 5	
Weight:	5%
Description:	Excel Midterm
ASSESSMENT ACTIVITIES 2	
Weight:	5%
Description:	Theory Midterm 1
ASSESSMENT ACTIVITIES 4	
Weight:	5%
Description:	Excel Training Certificate
ASSESSMENT ACTIVITIES 6	
Weight:	15%
Description:	Group Project

FINAL EXAM	
Minimum mark:	5
Weight:	60%
Description:	Theory Exam (20%) and Excel Practical Exam 80%)

5 RESOURCES

The use of mobile phones or any other electronic devices is not allowed in the classroom unless otherwise indicated by the teacher.

5.1 BIBLIOGRAPHY

1. From Data to Information to Intelligence-> -Holmes, D. E. (2017). Big data: a very short introduction. Oxford University Press. -O'Neil, C., & Schutt, R. (2013). Doing data science: Straight talk from the frontline. O'Reilly Media, Inc. -Runkler, T. A. (2020). Data analytics. Springer Fachmedien Wiesbaden. -Sherman, R. (2014). Business intelligence guidebook: From data integration to analytics. Newnes.

2. Building Data Teams -> -The Agile Manifesto: <https://agilemanifesto.org/> -Kanban: <https://www.atlassian.com/agile/kanban>

3. Data Collection -> -Three keys to building a data-driven strategy: <https://www.mckinsey.com/business-functions/mckinsey-digital/our-insights/three-keys-to-building-a-data-driven-strategy> -Famous errors in statistics: <https://www.statistics.com/famous-errors-in-statistics/> -Kaggle: <https://www.kaggle.com/datasets> -GitHub: <https://github.com/awesomedata/awesome-public-datasets> -Google Trends: <https://trends.google.com/trends/explore> -Structured vs. Unstructured Data: A Complete Guide: <https://www.talend.com/resources/structured-vs-unstructured-data/>

4. Ethical uses of Data -> -Derman, E. (2011). Models. Behaving. Badly.: Why Confusing Illusion with Reality Can Lead to Disaster, on Wall Street and in Life. Simon and Schuster. -Gigerenzer, G. (2008). Rationality for mortals: How people cope with uncertainty. Oxford University Press. -O'neil, C. (2016). Weapons of math destruction: How big data increases inequality and threatens democracy. Crown. -Richards, N. M., & King, J. H. (2014). Big data ethics. Wake Forest L. Rev., 49, 393. -Thaler, R. H., & Sunstein, C. R. (2008) Nudge: Improving decisions about health, wealth, and happiness. Yale University Press.

5. Types of Data Analytics -> -The 4 Types of Data Analytics: <https://www.kdnuggets.com/2017/07/4-types-data-analytics.html> -4 Types of Data Analytics and How to Apply Them: <https://www.michiganstateuniversityonline.com/resources/business-analytics/types-of-data-analytics-and-how-to-apply-them/> -Kalsbeek, R. (n.d.). Where to Start With the 4 Types of Analytics. <https://iterationinsights.com/>, from <https://iterationinsights.com/article/where-to-start-with-the-4-types-of-analytics/>

6. Exploratory Data Analysis and Descriptive Analytics -> -Descriptive Statistics: Expectations vs. Reality (Exploratory Data Analysis): <https://towardsdatascience.com/descriptive-statistics-expectations-vs-reality-exploratory-data-analysis-eda-8336b1d0c60b> -Bickel, P. J., Hammel, E. A., & O'Connell, J. W. (1975). Sex Bias in Graduate Admissions: Data from Berkeley. Science, 187(4175), 398-404. <https://doi.org/10.1126/science.187.4175.398> -Maydon, T. (n.d.). How To Dodge The Simpson's Paradox In Descriptive Analytics, from <https://insights.principa.co.za/the-simpsons-paradox-how-to-dodge-this-statistical-paradox-in-descriptive-analytics> -ParamitaGhosh. (2017, July 6). Fundamentals of Descriptive Analytics. DATAVERSITY. <https://www.dataversity.net/fundamentals-descriptive-analytics/> -Simpson's paradox. (2021). In Wikipedia. https://en.wikipedia.org/w/index.php?title=Simpson%27s_paradox&oldid=1037315783 -Anscombe's quartet. (2021). In Wikipedia. https://en.wikipedia.org/w/index.php?title=Anscombe%27s_quartet&oldid=1038700846

7. Excel training -> -Alexander, M., Kusleika, R., & Walkenbach, J. (2018). Excel 2019 Bible. John Wiley & Sons.
-Finance Chiefs Are Still Trying to Replace Excel With New Tools – WSJ:
<https://www.wsj.com/articles/finance-chiefs-are-still-trying-to-replace-excel-with-new-tools-11626859801>

5.2 SOFTWARE

NAME:	Zotero
DESCRIPTION OF USE:	Reference Tracking Software
AVAILABILITY:	Free
NAME:	Trello
DESCRIPTION OF USE:	Online Kanban Board
AVAILABILITY:	Free
NAME:	MS Excel
DESCRIPTION OF USE:	Spreadsheet Software
AVAILABILITY:	Available to students for free with Office 365 provided by CUNEF